

1.6 Classwork: Measurement review

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$

$$1 \text{ m} \approx 3.28 \text{ ft} \quad 1 \text{ km} = 1000 \text{ m} \quad 1 \text{ m} = 100 \text{ cm} \quad 1 \text{ cm} = 10 \text{ mm}$$

$$1 \text{ h} = 3600 \text{ s} \quad 1 \text{ yr} \approx 365.25 \text{ days} \quad 1 \text{ week} = 7 \text{ days}$$

1. Write down the number of significant digits in each measurement.

(a) 3.08 m

(c) 40.0 s

(e) 0.000 501 0 m

(b) 0.0062 kg

(d) 1.200 m

(f) $2.998 \times 10^8 \text{ m/s}$

2. Round each value to three significant figures. Copy the calculator display followed by three dots, then round: $\sqrt{2} = 1.4142135 \dots \approx 1.41$

(a) $\sqrt{11}$

(b) One mile, 1609.344 m

3. Write in scientific notation, rounded to three significant figures.

(a) The diameter of the Sun,
1,392,700 km

(b) The time light takes to cross a meter
stick, 0.000 000 003 335 64 s

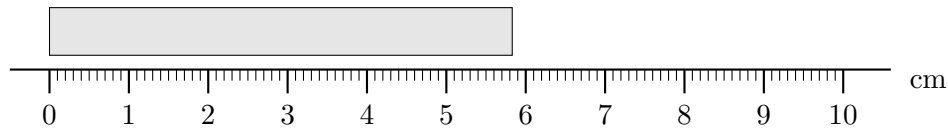
4. Write each value as an ordinary decimal number.

(a) The diameter of a red blood cell,
 $7.8 \times 10^{-6} \text{ m}$

(b) The distance from the Earth to the
Sun, $1.496 \times 10^8 \text{ km}$

5. Usain Bolt ran 100 m in 9.58 s. What was his average speed in kilometers per hour?

6. The bar below is measured with a centimeter ruler marked in millimeters.



- (a) Record the length of the bar. Include one estimated digit — the last digit, the one you are not certain of.
- (b) How many significant figures does your measurement have?
- (c) Write your measurement in meters.
7. Convert between units, rounding to three significant figures. Write each conversion as a single chain of fractions and show the units cancelling.
- (a) The Burj Khalifa is 828 meters tall. What is its height in feet?
- (b) The elevators in the Shanghai Tower climb at 20.5 meters per second. What is this speed in kilometers per hour?
8. Fingernails grow about 3.5 centimeters per year. How many millimeters does a fingernail grow in one week?